AFFICIONADO COFFEE ROASTERS – PRODUCT & REVENUE ANALYSIS

AFFICIONADO COFFEE ROASTERS – PRODUCT & REVENUE ANALYSIS

Project Submission Report

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Internship Program: Unified Mentor

Domain: Data Analytics

Tools Used: Python, Jupyter Notebook, Power BI

1. Introduction

In specialty coffee retail, understanding product-level performance is essential for improving business efficiency and maximizing profitability.

Afficionado Coffee Roasters handles a diverse menu of coffee, tea, bakery, and beverage products across multiple store locations. However, without structured analytical insights, identifying high-performing products and optimizing the menu becomes difficult.

This project focuses on analyzing transactional data to uncover product popularity, revenue contribution, category-level performance, and business optimization opportunities.

2. Problem Statement

Afficionado Coffee Roasters collects transaction-level sales data but lacks a systematic analytical framework to evaluate product performance.

This creates several challenges:

- Limited visibility into top-selling products
- Difficulty identifying low-performing menu items
- Lack of revenue contribution analysis
- Inefficient menu optimization decisions
- Limited strategic insight into store performance
- A data-driven product intelligence system is needed to improve decision-making.

3. Project Objectives

The primary objectives of this project are:

- Identify top-selling and least-selling products
- Analyze product-level revenue contribution
- Compare product popularity with profitability
- Evaluate category-wise revenue performance
- Measure store-wise revenue distribution
- Generate insights for menu optimization

4. Methodology

The project was performed using Python for data analysis and Power BI for dashboard development.

Step 1: Data Loading

Dataset imported into Jupyter Notebook using Pandas.

Step 2: Data Cleaning

- Data preprocessing included:
- Standardizing column names
- Validating missing values
- Checking data consistency

Step 3: Revenue Calculation

A new revenue column was created:

Revenue = Transaction Quantity × Unit Price

Step 4: Exploratory Data Analysis

- Analysis performed on:
- Product sales volume
- Revenue contribution
- Category performance
- Store performance

Step 5: Dashboard Development

Interactive Power BI dashboard created for visualization.

6. Python Analysis

The analysis was conducted using:

- Pandas
- Matplotlib
- NumPy

Key Analysis Performed

Product Sales Analysis

Identified top-selling and least-selling products.

Revenue Contribution Analysis

Measured revenue generated by each product.

Category-wise Revenue Analysis

Compared contribution of Coffee, Tea, Bakery, and Drinking Chocolate categories.

Store-wise Revenue Analysis

Evaluated revenue performance across store locations.

Product Efficiency Analysis

Calculated product efficiency using revenue per unit sold.

7. Dashboard Insights

The Power BI dashboard provides an interactive product intelligence system.

Dashboard Components

- KPI Cards
- Total Revenue

- Total Units Sold
- Top Product
- Top Store

Product Revenue Chart

Displays highest revenue-generating products.

Category Revenue Distribution

Visualizes category-level revenue contribution.

Store Revenue Distribution

Compares revenue across store locations.

Product Performance Summary

Provides actionable business insights.

Interactive Filters

Allows category, product type, and store-based analysis.

8. Key Findings

The analysis revealed important business insights.

1. Revenue Concentration

A small number of products contribute a large share of total revenue.

2. Popularity vs Profitability Difference

High-volume products are not always the highest revenue generators.

3. Category Performance Variation

Coffee products dominate revenue contribution.

4. Store-Level Revenue Differences

Revenue varies significantly across store locations.

5. Low-Performing Products

Several products contribute minimally and require evaluation.

9. Recommendations

Based on the analysis, the following recommendations are proposed.

Promote High-Revenue Products

Focus marketing efforts on top-performing products.

Review Low-Performing Products

Consider redesign, repositioning, or removal.

Optimize Inventory Planning

Prioritize stock management for high-demand products.

Improve Menu Structure

Reduce menu complexity to improve service efficiency.

Continuous Monitoring

Use dashboard analytics for regular product performance reviews.

10. Conclusion

This project successfully transformed raw transactional data into actionable business intelligence.

Using Python analysis and Power BI visualization, detailed insights were generated regarding product sales, revenue concentration, category performance, and store-level trends.

The findings provide strong support for data-driven menu optimization and strategic product planning for Afficionado Coffee Roasters.

This analytical framework enables smarter business decisions and long-term profitability improvement.

Tools Used

- Python
- Pandas
- Matplotlib
- Jupyter Notebook
- Power BI
- Microsoft Word

Future Scope

- Future enhancements may include:
- Predictive sales forecasting
- Customer segmentation analysis
- Recommendation systems
- Streamlit-based deployment
- Real-time dashboard integration